

Time-series Forecasting Baselines

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Let's change
perspective
again



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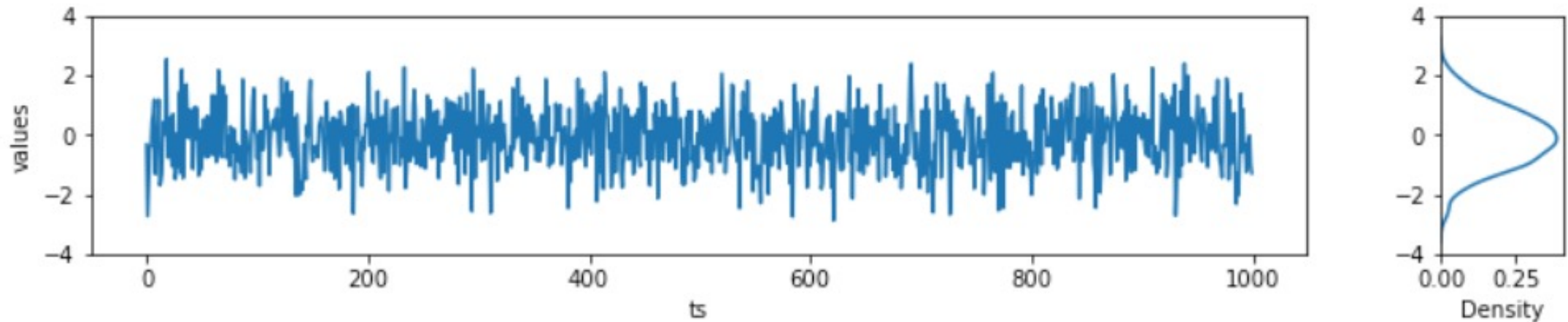
The story so far

- Very few time series in the wild are stationary
- Normally, they include a long-term trend (including aperiodic cycles) and one or more seasonal patterns
- We learned to
 - decompose a time series, and
 - test the stationarity of the residual



White noise: the stationary time series

- A sequence of random numbers sampled from a **Gaussian model** with zero mean and finite variance



- Notes:
 - different **realizations** of white noise are statistically described by the same mean (zero) and the same variance
 - the **mean** is the **best prediction**, and the **variance** indicates the prediction **error**

Fact

If a time series is stationary,... ... it is predictable



Thus, it is possible to **forecast** a time series with this characteristics



Thinking

- How far can we predict?
 - short-term
 - A **one-step-ahead** is a forecast for the next observation only
 - medium- and long-term (no line of demarcation)
 - A **multi-step-ahead** forecast is for 2,3,..., n steps ahead
- What can we predict?
 - Trend: long-term, the easiest to predict
 - Seasonal repetition: medium-term, does it really repeat identically?
 - Residual: short-term, the harder to predict

Let's build the
intuition of
forecasting



The overall forecasting process

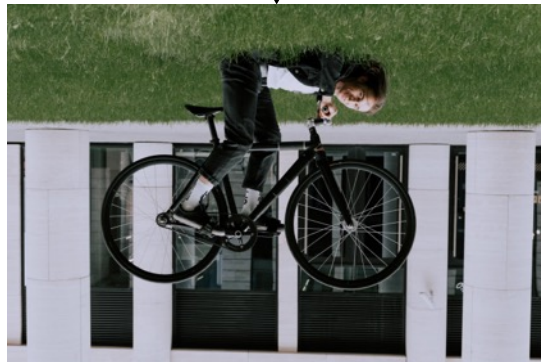
original time series



residual component



recomposed time series



decomposed time series



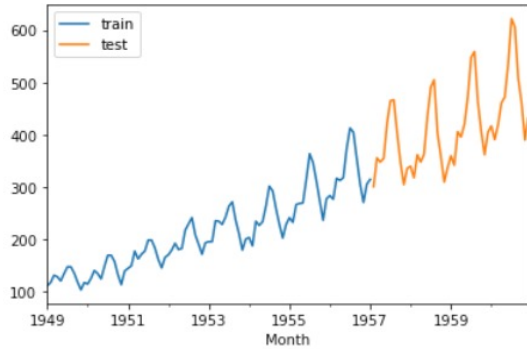
prediction of the residual



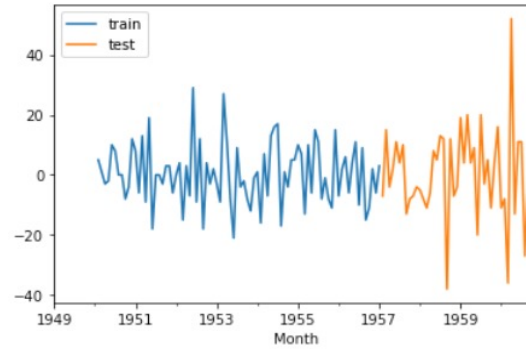
predicted time series

The overall forecasting process

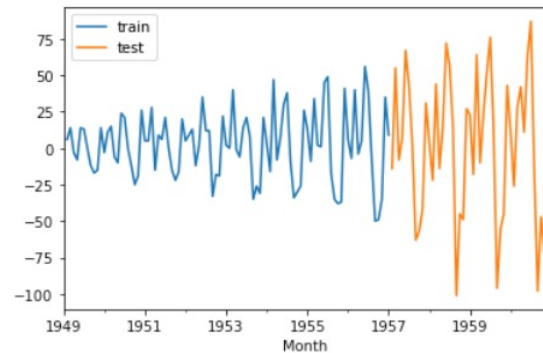
original time series



residual component



recomposed time series



decomposed time series



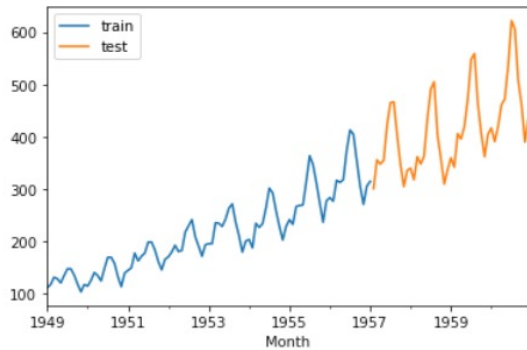
prediction of the residual



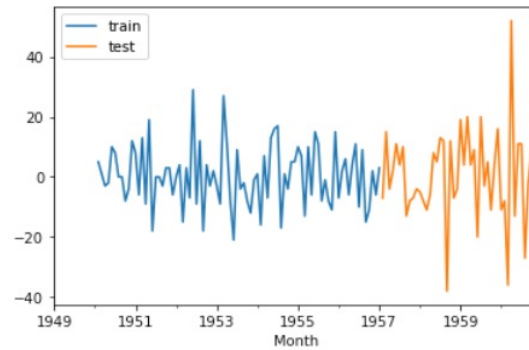
predicted time series

The overall forecasting process

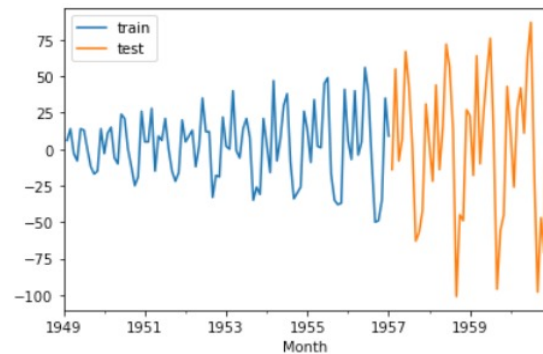
original time series



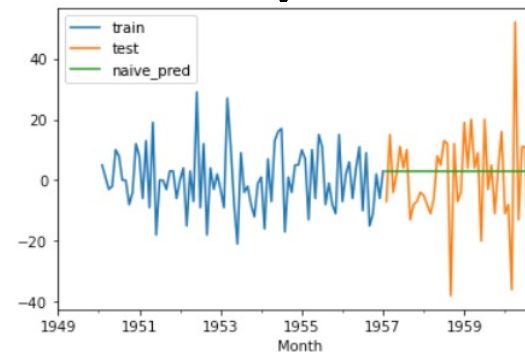
residual component



recomposed time series



decomposed time series



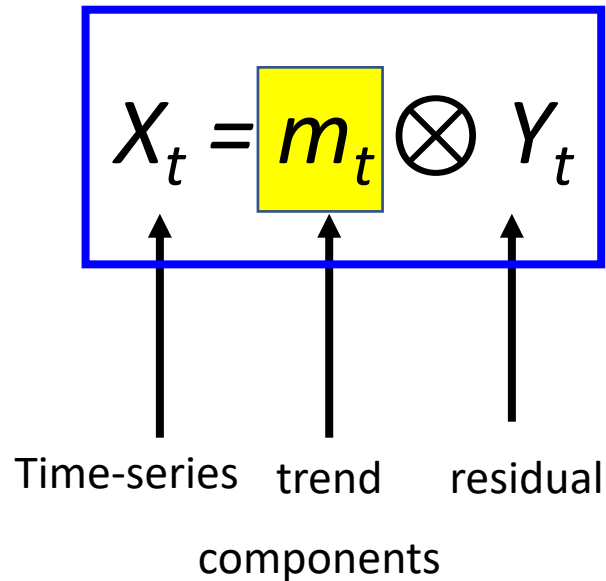
prediction of the residual



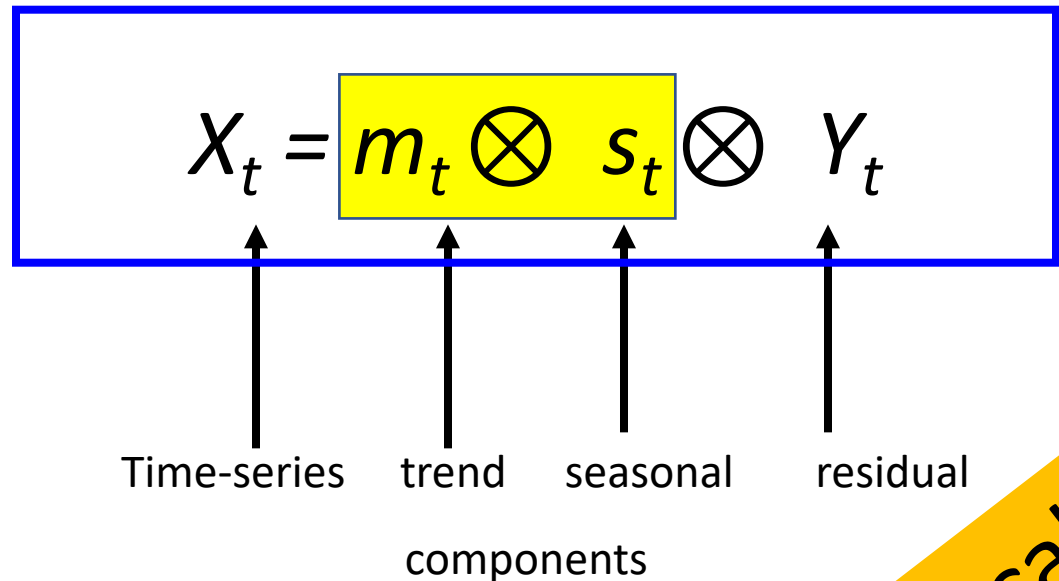
predicted time series

Two simplified time series models

Non-seasonal Decomposition Model with Trend

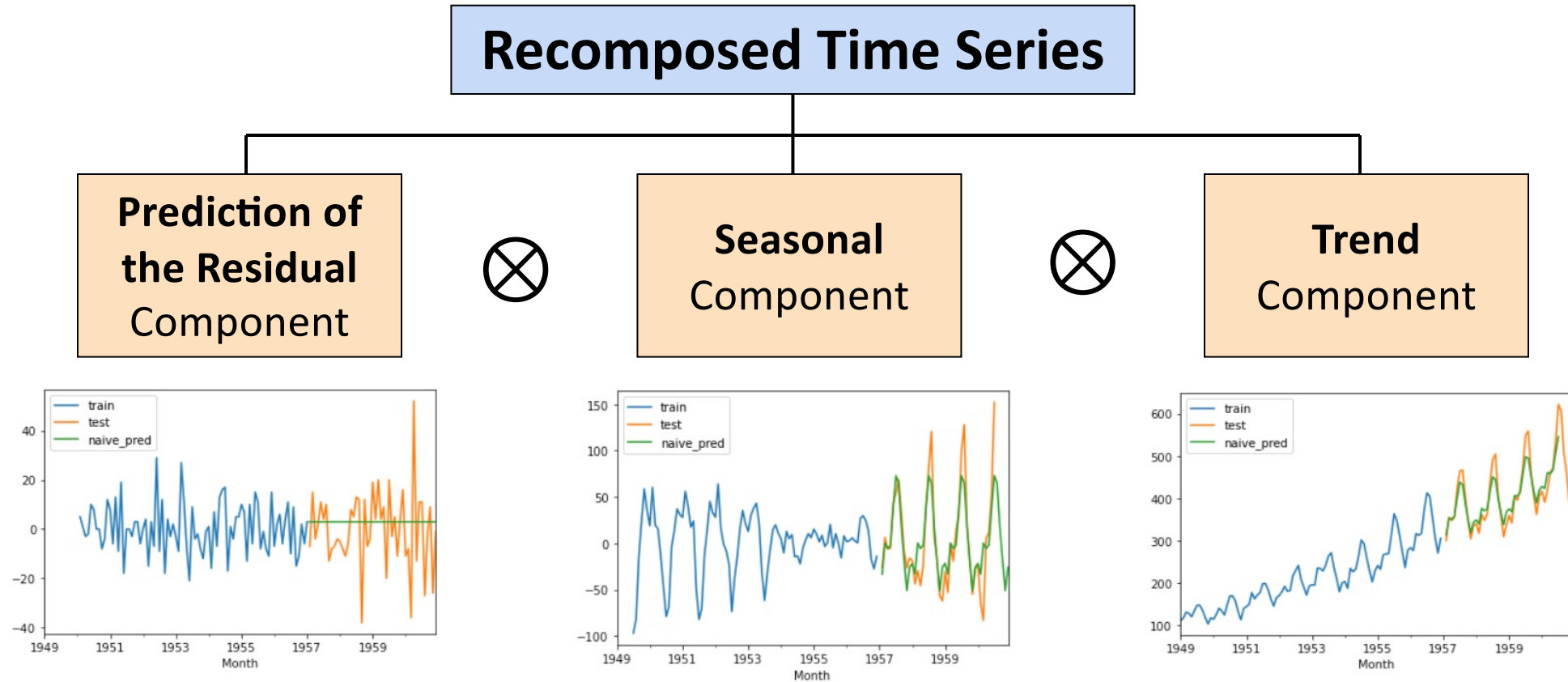


Decomposition Model **with** Trend and **Seasonal** Components



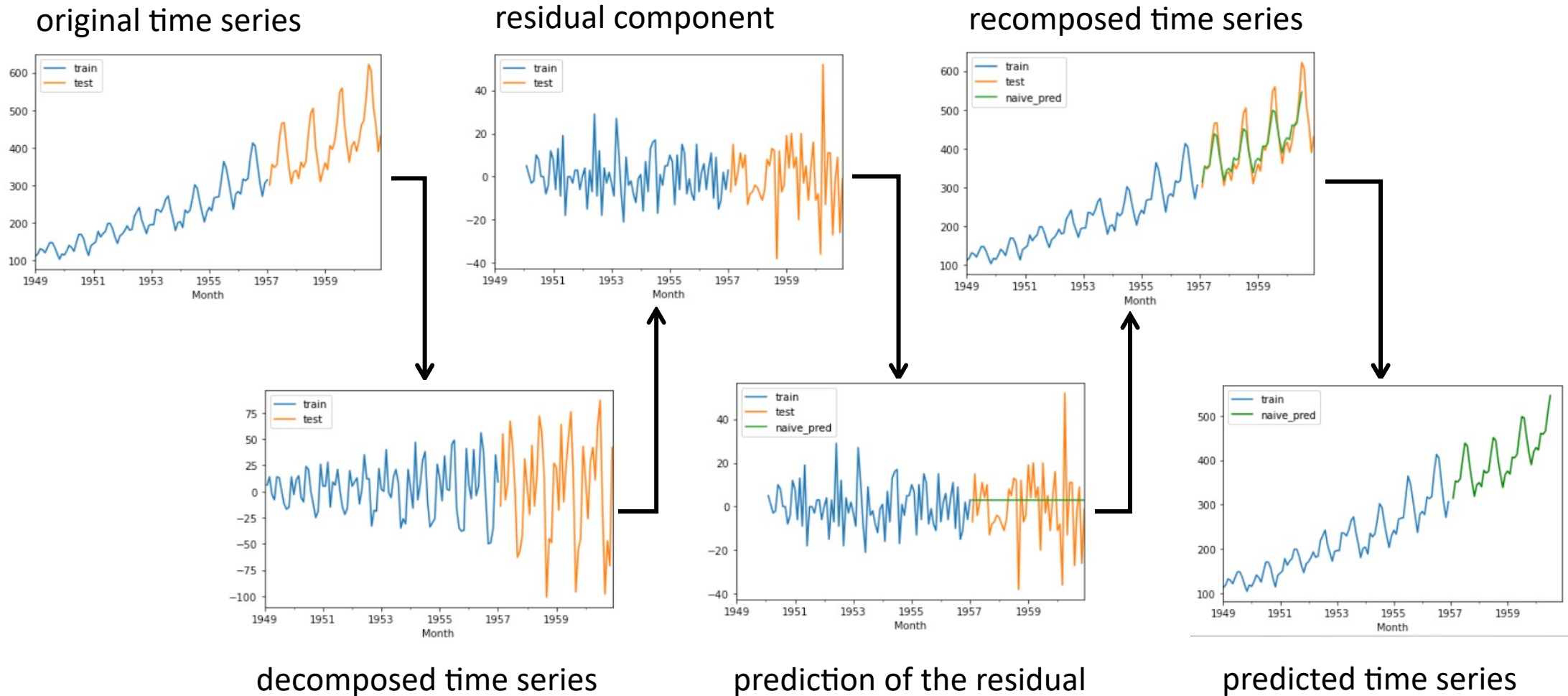
Recall

We can recompose the time series



$\otimes$ This character is a placeholder for various mathematical operation used to assemble the components

The overall forecasting process



Formally...



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How do we measure accuracy?

- mean absolute error (MAE)

$$MAE = \sum_i \frac{|Y_i - \hat{Y}_i|}{n}$$

- mean absolute percent error (MAPE)

$$MAPE = \frac{100}{n} \sum_{i=1}^n \frac{|Y_i - \hat{Y}_i|}{Y_i}$$

- mean square error (MSE)

$$MSE = \sum_{i=1}^n \frac{(Y_i - \hat{Y}_i)^2}{n}$$

- root mean square error (RMSE)

$$RMSE = \sqrt{MSE}$$

Which one to use, why and when?

Error Metric	Benefit	Limitation
Mean Absolute Error (MAE)	Easy to calculate, understand and interpret.	Sensitive to outliers; may not penalize large errors sufficiently.
Mean Absolute Percentage Error (MAPE)	Provides a percentage error representation; can be used to make comparisons across different datasets.	Susceptible to division by zero, i.e., when actual values are close to zero. Biased towards actual values. Asymmetric.
Mean Square Error (MSE)	Penalizes larger errors more significantly than small errors	Units are squared, which might be less intuitive in some contexts.
Root Mean Square Error (RMSE)	Provides an interpretable scale for MSE like the original data.	Sensitive to outliers like MSE; not robust against extreme values.

Example 1

A demand forecasting model that predicts the demand for a particular product

✓ MAE

⚠ RMSE

✓ MAPE

✗ MSE

Example 1 - Solution

MAPE

- PROS:
 - Allows accuracy comparison across different products
 - Easily communicable to business ("5% error")
 - Intuitive for non-technical stakeholders
- WHEN TO USE:
 - Demand is always significantly greater than zero
 - A percentage metric is needed for comparisons

RMSE

- CONSIDERATIONS:
 - Can be useful if large errors are particularly costly
 - Less intuitive for business communication
- WHEN TO AVOID:
 - If outliers are not critical for the business
 - If communication with non-technical stakeholders is a priority

MAE

- PROS:
 - Directly interpretable in product units
 - Not distorted by extreme values
- WHEN TO USE:
 - You want to know the average error in product units
 - Error scale is important for planning

MSE

- WHY NOT:
 - Squared unit of measurement difficult to interpret
 - Less useful for practical business decisions
 - RMSE would be preferable if penalizing large errors is needed

Example 2

A traffic forecasting model that predicts the number of vehicles on a particular road

✓ MAE

✓ RMSE

⚠ MAPE

✗ MSE

Example 2 - Solution

MAPE ⚠️

- CONSIDERATIONS:
 - Problematic during night hours with low traffic (values close to zero)
 - Can give distorted results when traffic is highly variable
- WHEN TO AVOID:
 - During hours with very low traffic
 - When comparing roads with very different traffic volumes

RMSE ✅

- PROS:
 - Appropriately penalizes large errors, critical in traffic
 - Particularly useful for identifying unexpected congestions
 - Maintains the unit of measurement (number of vehicles)
- WHEN TO USE:
 - When identifying unexpected traffic peaks is important
 - When large errors can lead to significant problems

MAE ✅

- PROS:
 - Provides a direct estimate of the average error in number of vehicles
 - More robust to outliers compared to RMSE
 - Easy to communicate to stakeholders
- WHEN TO USE:
 - For a general accuracy assessment
 - When you want a more stable metric over time

MSE ❌

- WHY NOT:
 - The squared unit of measurement (vehicles²) has no practical meaning
 - RMSE is preferable if you want to penalize large errors
 - Too sensitive to outliers for this context

Example 3

A weather forecasting model that predicts the temperature for a particular location

 MAE

 RMSE

 MAPE

 MSE

Example 3 - Solution

MAPE ❌

- WHY NOT:
 - **Problematic with temperatures near zero (division issues)**
 - Different meanings for Celsius and Fahrenheit scales
 - Can give misleading results with temperature data
 - Percentages are not natural for temperature differences

RMSE ✅

- PROS:
 - Captures the impact of larger temperature errors
 - Useful for identifying significant forecasting failures
 - Still maintains temperature units
- WHEN TO USE:
 - When large errors are more critical (e.g., extreme weather events)
 - For technical evaluation of the model
 - When variance in errors is important

MAE ✅

- PROS:
 - Directly interpretable in temperature units (°C or °F)
 - Provides average error magnitude that matters to users
 - Not overly sensitive to occasional large misses
 - Most intuitive ("off by 2 degrees on average")
- WHEN TO USE:
 - For general forecast accuracy reporting
 - When all temperature errors are equally important

MSE ❌

- WHY NOT:
 - Squared temperature units are not meaningful
 - Less interpretable than RMSE
 - Not suitable for public communication
 - No advantage over RMSE in this context

Let's focus on predicting stationary time-series

original time series



residual component



recomposed time series



decomposed time series



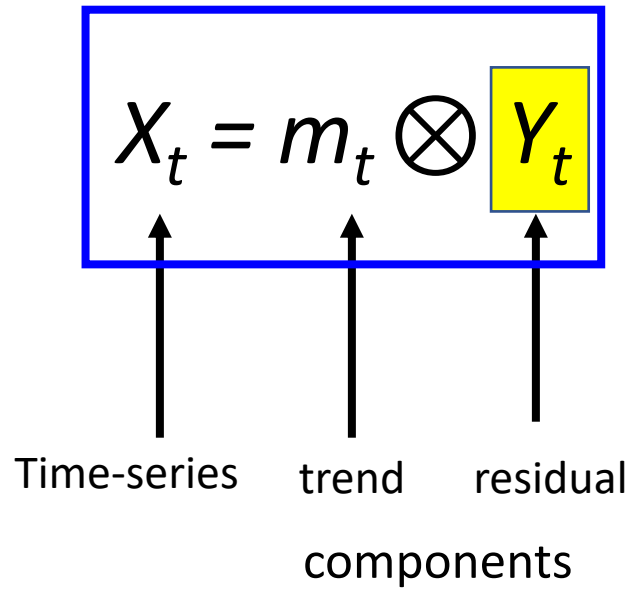
prediction of the residual



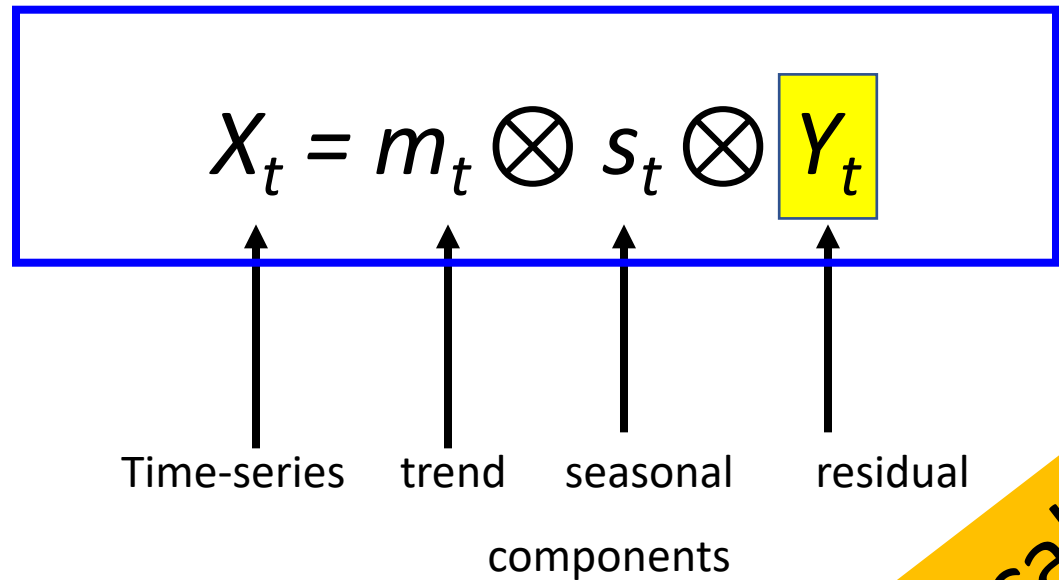
predicted time series

Two simplified time series models

Non-seasonal Decomposition Model with Trend



Decomposition Model **with** Trend and **Seasonal** Components



Recall

Base line forecasting methods

“Naïve” approach

- The only important value to make predictions is the last one

$$\hat{y}_{t+1} = y_t$$

“Average” approach

- All the previous values are equally important to make predictions

$$\hat{y}_{t+1} = \frac{1}{N} \sum y_i$$

Base line forecasting methods

1. Load the airline time series
2. Make it stationary as in the previous lab
3. Predict using both the “Naïve” approach and the “Average” approach
4. Compute the accuracy metrics for both predictions
5. Which prediction is better?



More sophisticated forecasting methods

Last-k average approach

- Only the Last-k previous values are important to make predictions

$$\hat{y}_{t+1} = \frac{1}{k} \sum_{i=0}^{k-1} y_{t-i}$$

- PROBLEM: how do we choose k?

Last-k average forecasting methods

Continuing from previous

1. Predict using “Last-k average” approach
2. Which is the best k ?
3. Does the “Last-k average” approach outperform the two previous one for at least a value of k ?



Thinking...

- The last-k appears to be in the middle way between the «Naive» approach and «Average» approach
- but it is hard to find a k that clearly separates
 - the data to “forget” (older than k)
 - from those to use
- Is there any other way to consider all the values and still introduce a notion of “forgetting”?
- Are all values equally important?

More sophisticated forecasting methods

Exponential Smoothing

- A middle way between Naive approach and Average approach:
 - Not only the last one (Naive)
 - Not all equally important (Average)

$$\hat{y}_{t+1} = \alpha y_t + (1 - \alpha) \hat{y}_t$$

Where $0 < \alpha < 1$ is the smoothing level

- Notes:
 - **α close to 1** indicates **fast forgetting** (only the most recent values influence the forecasts)
 - **α close to 0** indicates **slow forgetting** (past observations have a large influence on forecasts)
- Often it easier to set α as the number of points (*span*) with a considerable impact

$$\alpha = 2/(\text{span} + 1)$$

More sophisticated forecasting methods

Why is it call exponential smoothing?

- Starting with $\hat{y}_{t+1} = \alpha y_t + (1 - \alpha)\hat{y}_t$ we can substitute for $\hat{y}_t$ and get

$$\begin{aligned}\hat{y}_{t+1} &= \alpha y_t + (1 - \alpha)[\alpha y_{t-1} + (1 - \alpha)\hat{y}_{t-1}] \\ &= \alpha y_t + \alpha(1 - \alpha)y_{t-1} + (1 - \alpha)^2\hat{y}_{t-1}\end{aligned}$$

Continuing to substitute until the first element of the time-series leads to

$$\hat{y}_{t+1} = \alpha y_t + \alpha(1 - \alpha)y_{t-1} + \alpha(1 - \alpha)^2y_{t-2} + \dots + \alpha(1 - \alpha)^{t-1}y_1 + (1 - \alpha)^t l_0$$

where l_0 represents the first fitted values at time 1 (the process has to start somewhere)
Hence, the forecasts are weighted averages of past observations, with **the weights decaying exponentially as the observations get older.**

More sophisticated forecasting methods

Exponential Smoothing is a streaming algorithm

$$\begin{aligned}\hat{y}_{t+1} &= \alpha y_t + (1 - \alpha)\hat{y}_t \\ &= \alpha y_t + \hat{y}_t - \alpha \hat{y}_t \\ &= \hat{y}_t + \alpha y_t - \alpha \hat{y}_t \\ &= \hat{y}_t + \alpha(y_t - \hat{y}_t) \\ &= \hat{y}_t + \alpha(e_t)\end{aligned}$$

- The **next prediction** is the sum of the **current prediction** and **alpha times the current error** (i.e., the error of the current prediction)

Exponential smoothing forecasting methods

Continuing from previous

1. Predict using “exponential smoothing” approach
2. Which is the best smoothing level?
3. Does the “exponential smoothing” approach outperform the three previous ones for at least a value of the smoothing level?



(-: we can predict a stationary time series :-)

original time series



residual component



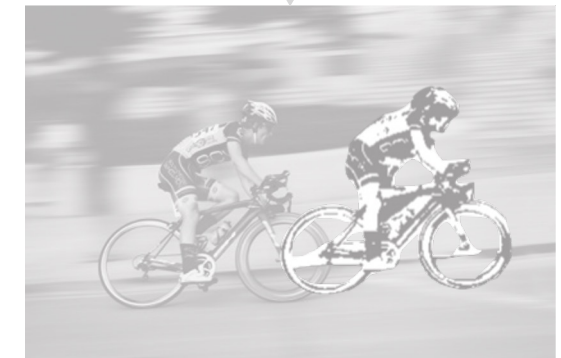
recomposed time series



decomposed time series



prediction of the residual



predicted time series

What about directly forecasting a time series with trend and seasonality?

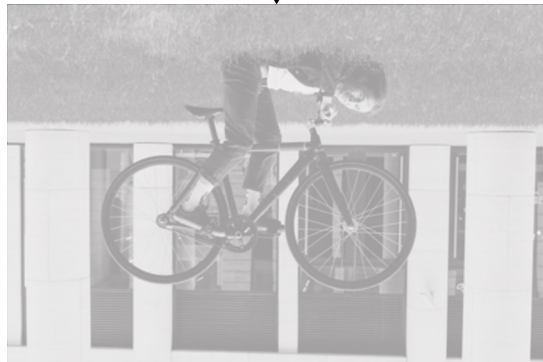
original time series



residual component



recomposed time series



decomposed time series



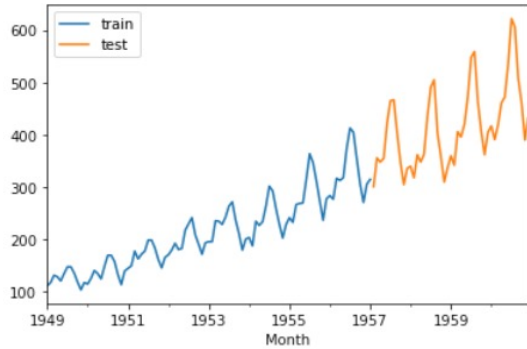
prediction of the residual



predicted time series

Forecasting a time series with trend using the Holt method

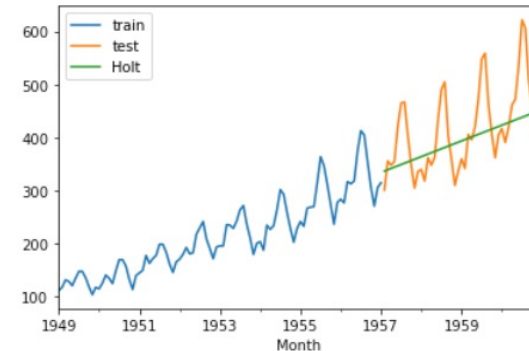
original time series



residual component



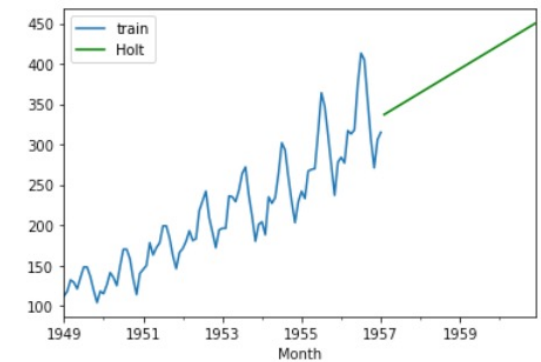
recomposed time series



decomposed time series



prediction of the residual



predicted time series

More sophisticated forecasting methods

Double Exponential Smoothing (a.k.a., Holt's linear method)

- An extension to Exponential Smoothing that **adds support for trends** using an additional smoothing factor (called β) to control the decay of the influence of the change in trend
- The method supports trends that change in different ways:
 - **additive** when the trend is **linear**
 - Forecast equation $\hat{y}_{t+h} = l_t + hb_t$
 - Level equation $l_t = \alpha y_t + (1 - \alpha)(l_{t-1} + b_{t-1})$
 - Trend equation $b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}$
 - **multiplicative** when the trend is exponential
 - Forecast equation $\hat{y}_{t+h} = l_t(b_t)^h$

More sophisticated forecasting methods

Double Exponential Smoothing (a.k.a., Holt's linear method)

- Zooming a bit more on the equations

- Forecast equation
- Level equation
- Trend equation

$$\hat{y}_{t+h} = l_t + hb_t$$

$$l_t = \alpha y_t + (1 - \alpha)(l_{t-1} + b_{t-1})$$

$$b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}$$

The simple exponential smoothing equation

Adding a trend

The trend gets updated based on the most recent error

More sophisticated forecasting methods

Double Exponential Smoothing (a.k.a., Holt's linear method)

- Conclusions
 - The trend can vary adaptively over time
 - The trend smoothing parameter β controls the speed of adjusting the trend

Double Exponential smoothing forecasting methods

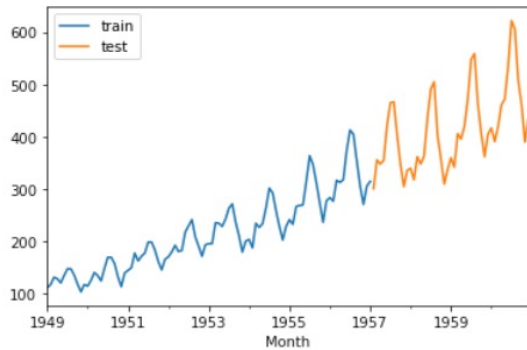
Continuing from previous

1. Pick the original data without detrending and deseasonizing them
2. Predict the test using the “double exponential smoothing” approach
3. Which are the best smoothing levels?



Forecasting a time series with trend and seasonality using the Holt-Winters method

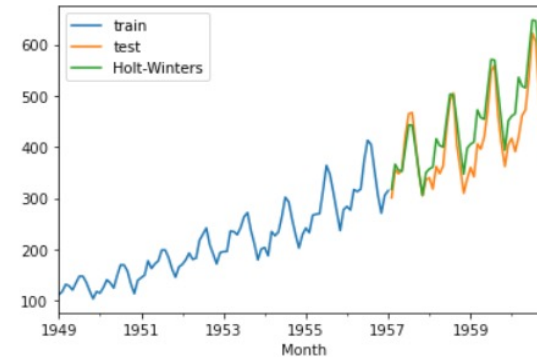
original time series



residual component



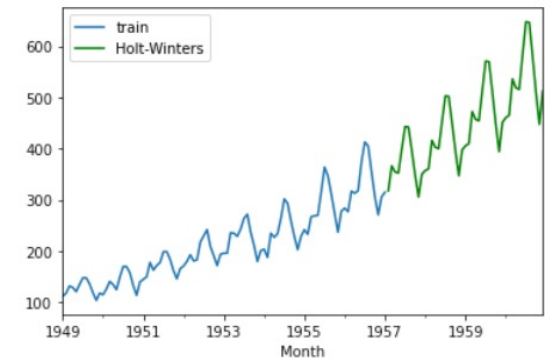
recomposed time series



decomposed time series



prediction of the residual



predicted time series

More sophisticated forecasting methods

Triple Exponential Smoothing (a.k.a. Holt-Winters method)

- An extension to Exponential Smoothing that **adds support for** seasonality using a new parameter *gamma* that controls the influence on the seasonal component
- As with the trend, the seasonality may be modelled as an **additive or multiplicative** process for a linear or exponential change in the seasonality.
- Additive equations

• Forecast equation	$\hat{y}_{t+h} = l_t + hb_t + s_{t+h-d}$
• Level equation	$l_t = \alpha(y_t - s_{t-d}) + (1 - \alpha)(l_{t-1} + b_{t-1})$
• Trend equation	$b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}$
• Seasonality equation	$s_t = \gamma(y_t - l_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-d}$

More sophisticated forecasting methods

Triple Exponential Smoothing (a.k.a. Holt-Winters method)

- Zooming a bit more on the equations

- Forecast equation
- Level equation
- Trend equation
- Seasonality equation

$$\hat{y}_{t+h} = l_t + hb_t + s_{t+h-d}$$

$$l_t = \alpha(y_t - s_{t-d}) + (1 - \alpha)(l_{t-1} + b_{t-1})$$

$$b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}$$

$$s_t = \gamma(y_t - l_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-d}$$

Here we deseasonalise!

Here we detrend!

Same as in Double Exponential Smoothing

More sophisticated forecasting methods

Triple Exponential Smoothing (a.k.a. Holt-Winters method)

- Conclusions

- Both the trend and the seasonality can vary adaptively over time
- The trend smoothing level β and the seasonality smoothing level γ control the speed of adjusting the trend and the seasonality, respectively
- The only fixed parameter is the period d of the seasonality

Triple Exponential smoothing forecasting methods

Continuing from previous

1. Pick the original data without detrending and deseasonizing them
2. Predict the test using the “triple exponential smoothing” approach
3. Which are the best smoothing levels?
4. Does the “triple exponential smoothing” approach outperform the previous ones?



Time-series Forecasting Baselines

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